

# Impact of Input Membership Function Design on Fuzzy-Adaptive Particle Swarm Optimization: An Empirical Sensitivity Analysis

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**Abstract**—This study investigates the impact of input membership function design on the performance of a fuzzy-controlled Particle Swarm Optimization (PSO). Although fuzzy PSO research has typically adopted input partitions without systematic comparison, this work isolates the effect of varying the linguistic representation of iteration progress and population diversity by holding the output set and rule base constant. Four input set geometries crossed with two granularity levels are evaluated on six benchmark functions spanning unimodal and multimodal problem types. Results show a problem-type-dependent pattern: on unimodal functions the baseline PSO with linear inertia weight outperforms all fuzzy variants, whereas on the multimodal problems tested the fuzzy controller improves mean fitness by up to 22.9% and reduces variability by up to 63%. Within the fuzzy-controlled variants, the choice of input partition geometry measurably affects performance, with moderate-overlap configurations and three-label granularity yielding the most robust results across the benchmark suite.

**Index Terms**—Particle Swarm Optimization, fuzzy logic, inertia weight, membership functions, linguistic granularity, adaptive parameter control

## I. INTRODUCTION

Particle Swarm Optimization (PSO) is a population-based metaheuristic whose convergence behavior depends heavily on how global exploration and local exploitation are balanced [?]. The inertia weight is the main lever for controlling that balance, which has motivated a long line of adaptive strategies [?].

Fuzzy control systems offer a way to adapt PSO parameters online through qualitative descriptions of the search state,

and have been used as an alternative to deterministic or iteration-based schedules [?]. In practice, however, most fuzzy-controlled PSO variants adopt fixed linguistic partitions and predefined membership functions chosen a priori. How these partitions are shaped is rarely questioned.

Consequently, the effect of linguistic granularity and membership function configuration on PSO performance has not been systematically studied. It remains unclear to what extent different fuzzy linguistic schemes, even under identical control logic, influence optimization behavior across problems with different landscape characteristics.

This paper presents an experimental analysis of fuzzy-controlled PSO variants that differ exclusively in linguistic granularity and membership function design. No new optimization algorithm is proposed. Instead, the study compares alternative fuzzy semantic configurations under identical control logic and measures their effect on performance and robustness across representative benchmark functions. The results show that fuzzy scheme design is a non-trivial experimental factor in adaptive metaheuristic optimization, one that prior work has largely treated as a fixed choice.

The rest of the paper proceeds as follows. Section ?? reviews relevant work on fuzzy-adaptive parameter control in PSO and related optimization frameworks. Section ?? describes the fuzzy-controlled PSO framework and the linguistic configurations evaluated in this study. Section ?? details the benchmark functions, parameter settings, and experimental protocol. The experimental results and their discussion are

presented in Section ?? . Finally, Section ?? summarizes the main findings and outlines directions for future research.

## II. RELATED WORK

Two lines of prior work are relevant here: strategies for adaptive inertia weight control in PSO—particularly those based on fuzzy logic—and studies on how membership function design and linguistic granularity affect fuzzy system behavior.

### A. Adaptive Inertia Weight Control in PSO

Since Shi and Eberhart introduced the inertia weight [?], many authors have tried to replace the original linear decreasing schedule with state-dependent alternatives. Chatterjee and Siarry [?] introduced nonlinear decreasing functions that accelerate convergence in later iterations, while Nickabadi et al. [?] designed a feedback-based mechanism that adjusts  $w$  according to the swarm’s success ratio, coupling the control signal to actual search progress rather than to a predefined temporal schedule.

Fuzzy logic provides a qualitative alternative by encoding adaptation knowledge as linguistic rules. Shi and Eberhart [?] first showed that a Mamdani-type controller driven by swarm performance indicators can outperform static schedules. Their work set the stage for a line of research in which  $w$  is governed by linguistic inference rather than by an explicit function of the iteration count. Valdez et al. [?] extended this approach with co-evolutionary strategies and showed consistent advantages over classical PSO across diverse benchmarks. Nobile et al. [?] proposed a self-tuning mechanism that simultaneously adjusts multiple PSO parameters at the particle level, while Xia et al. [?] introduced a multiple-input multiple-output fuzzy controller for adapting learning coefficients and elite selection. Olivas et al. [?] showed that interval type-2 controllers can handle additional uncertainty from noisy landscapes, and Roy et al. [?] recently proposed evolving fuzzy policies that adapt up to seven hyperparameters—arguably the broadest fuzzy control architecture for PSO published so far.

A common trait in the works reviewed above is that the input membership functions—defining how diversity, fitness progress, or other indicators are fuzzified—are chosen a priori, and their geometric configuration is seldom examined as an experimental factor. Research efforts have concentrated on rule base design, output set selection, or the number of controlled parameters, while the input partition tends to be treated as a fixed design element.

### B. Membership Function Design and Linguistic Granularity

Work in the broader fuzzy systems community already points in this direction. Olivas et al. [?] demonstrated that different membership function shapes produce statistically significant performance differences even when the rule base remains unchanged, which implies that membership functions carry semantic content beyond their mathematical form. Metaheuristic-based tuning of membership function parameters reinforces this point, having shown sensitivity across several application domains [?], [?].

At the partition level, Cordon et al. [?] established guidelines for obtaining uniform fuzzy partitions with adequate granularity, showing that properties such as overlap degree and label spacing are constrained by consistency requirements. Guillaume and Charnomordic [?] proposed methods for generating interpretable partitions from data while preserving semantic coherence. Taken together, these works show that the geometric configuration of membership functions—support width, overlap, and boundary treatment—has a direct effect on both interpretability and control behavior.

A second design axis is the number of linguistic labels used to partition the input space, commonly called linguistic granularity. Castiello et al. [?] showed that adaptive granularity can improve discrimination in classification tasks without sacrificing interpretability. In the context of optimization, Huang et al. [?] and Zhang et al. [?] showed that the granularity of linguistic scales affects convergence behavior and solution quality in PSO-related decision-making problems.

Despite this evidence, the specific influence of input membership function geometry and linguistic granularity on fuzzy-controlled PSO has received little dedicated attention. Most fuzzy PSO studies in the literature adopt a single partition—typically three uniformly spaced triangular labels—without comparing it against alternatives. The present work contributes to filling this gap through a factorial design in which four input set geometries, differing in overlap degree and boundary treatment, are crossed with two granularity levels (3 and 5 labels) under a constant output set and rule base. Because everything else is held fixed, differences in performance can be attributed to the input partition itself.

## III. FUZZY-CONTROLLED PSO FRAMEWORK

This section describes the PSO framework and the fuzzy control layer added to it. A swarm-level fuzzy inference system adapts the inertia weight at every iteration. The question under study is whether—and how much—the geometric configuration of the *input* membership functions affects the resulting optimization behavior. To answer it, the output fuzzy set and rule base are kept the same across all variants; only the input partition changes.

### A. Baseline Particle Swarm Optimization

The baseline corresponds to the canonical PSO formulation, where particle motion is governed by inertia, cognitive attraction toward the personal best, and social attraction toward the global best position. Particle dynamics are defined by

$$v_i(t+1) = w \cdot v_i(t) + c_1 r_1 (p_i - x_i(t)) + c_2 r_2 (g - x_i(t)), \quad (1)$$

$$x_i(t+1) = x_i(t) + v_i(t+1), \quad (2)$$

where  $p_i$  and  $g$  denote the personal and global best positions, and  $r_1, r_2 \in [0, 1]$  are uniformly distributed random variables.

In the baseline configuration, the inertia weight follows a deterministic linear decreasing schedule from 0.9 to 0.1 over the optimization horizon [?]. The acceleration coefficients are set to  $c_1 = c_2 = 2.0$  for all configurations. This baseline is the

reference against which fuzzy-controlled inertia adaptation is measured.

### B. Fuzzy Control System for Inertia Weight Adaptation

In the fuzzy-controlled variants, a Mamdani-type fuzzy inference system (FIS) is embedded into the PSO loop and evaluated once per iteration at the swarm level. At iteration  $t$ , the controller outputs an inertia weight  $w_t$  based on two normalized indicators: iteration progress and population diversity.

Iteration progress is defined as

$$t_{\text{progress}} = \frac{t}{T_{\text{max}}}, \quad (3)$$

where  $T_{\text{max}}$  is the maximum number of iterations. Population diversity measures the normalized spatial dispersion of the swarm:

$$D(t) = \frac{1}{d} \sum_{k=1}^d \frac{\max_i x_{i,k}(t) - \min_i x_{i,k}(t)}{U_k - L_k}, \quad (4)$$

where  $d$  is the problem dimensionality and  $[L_k, U_k]$  are the bounds of dimension  $k$ .

The fuzzy controller maps these inputs to an inertia weight constrained to  $[w_{\min}, w_{\max}]$ . Defuzzification is performed using the centroid method:

$$w_t = \frac{\int z \mu_{\text{agg}}(z) dz}{\int \mu_{\text{agg}}(z) dz}, \quad (5)$$

where  $\mu_{\text{agg}}(z)$  denotes the aggregated output membership function. Fig. ?? illustrates the integration of the fuzzy controller within the PSO iteration loop.

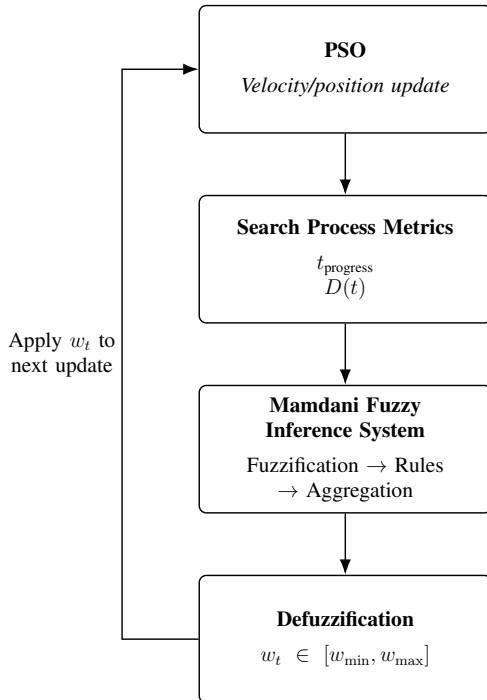


Fig. 1. Architecture of the Fuzzy-Controlled PSO Framework

### C. Input Membership Function Configurations

The central factor under study is the geometric configuration of the input membership functions—the partitions applied to diversity ratio and iteration progress. While prior work on fuzzy-controlled PSO has explored diverse rule bases, multi-parameter tuning, and alternative controller architectures [?], [?], [?], the sensitivity of the controller to input partition geometry has received limited attention. This work addresses that gap by evaluating four input set configurations, each employing triangular membership functions with different overlap and boundary characteristics.

All four configurations share the same linguistic labels—low, medium, and high for diversity; early, mid, and late for iteration progress—and are applied identically to both input variables. Any difference in controller behavior therefore comes from the shape of the partitions, not from a change in the underlying semantic structure.

- **I1 (Standard):** Uniformly distributed triangular functions with moderate overlap. This is the textbook partition and serves as the default reference.
- **I2 (Narrow):** Reduced support width and zero overlap between adjacent labels. Transitions between linguistic regions are sharper, so the controller switches rules more abruptly.
- **I3 (Wide):** Increased support width with substantial overlap between neighbors. The result is a smoother, more gradual inertia adaptation.
- **I4 (Shoulder):** Shoulder functions at the extremes and a standard triangle at the center. Near domain boundaries the extreme label stays fully active ( $\mu = 1$ ), which locks a single rule in place at extreme input values.

Two levels of linguistic granularity are also considered: a three-label partition (low, medium, high) and a five-label one (very\_low, low, medium, high, very\_high). In the five-label case, each input set extends its geometric principles—overlap degree, boundary treatment—to five uniformly spaced labels. Crossing four input geometries with two granularity levels yields eight fuzzy-controlled PSO variants.

Tables ?? and ?? detail the triangular parameters  $(a, b, c)$  for each configuration at both granularity levels. The same partition is applied symmetrically to both input variables.

### D. Output Fuzzy Set Configuration

The output variable is the inertia weight  $w$ , defined over  $[0, 1]$  and then scaled to  $[w_{\min}, w_{\max}]$ . A single output set (Set A) is shared by all variants so that the output side does not confound the comparison of input partitions.

Set A employs uniformly distributed triangular membership functions with symmetric overlap. For the three-label case, the output labels are low  $(0.00, 0.25, 0.5)$ , medium  $(0.25, 0.50, 0.75)$ , and high  $(0.50, 0.75, 1.0)$ . The five-label extension subdivides the same domain into very\_low, low, medium, high, and very\_high with equal spacing.

TABLE I

THREE-LABEL TRIANGULAR MF PARAMETERS ( $a, b, c$ ) FOR EACH INPUT SET. IN I4, EXTREME LABELS USE SHOULDER FUNCTIONS WHERE  $a=b$  OR  $b=c$ .

| Label      | I1 (Standard)   | I2 (Narrow)       | I3 (Wide)         | I4 (Shoulder)    |
|------------|-----------------|-------------------|-------------------|------------------|
| Low/Early  | (0.0, 0.2, 0.4) | (0.0, 0.16, 0.33) | (0.0, 0.25, 0.5)  | (0.0, 0.0, 0.35) |
| Medium/Mid | (0.3, 0.5, 0.7) | (0.33, 0.5, 0.66) | (0.25, 0.5, 0.75) | (0.2, 0.5, 0.8)  |
| High/Late  | (0.6, 0.8, 1.0) | (0.66, 0.84, 1.0) | (0.5, 0.75, 1.0)  | (0.65, 1.0, 1.0) |

TABLE II

FIVE-LABEL TRIANGULAR MF PARAMETERS ( $a, b, c$ ) FOR EACH INPUT SET. I4 RETAINS SHOULDER FUNCTIONS AT EXTREMES; I2 ACHIEVES ZERO OVERLAP BETWEEN ADJACENT LABELS.

| Label               | I1 (Standard)     | I2 (Narrow)     | I3 (Wide)         | I4 (Shoulder)     |
|---------------------|-------------------|-----------------|-------------------|-------------------|
| Very Low/Very Early | (0.0, 0.15, 0.3)  | (0.0, 0.1, 0.2) | (0.0, 0.15, 0.35) | (0.0, 0.0, 0.25)  |
| Low/Early           | (0.2, 0.35, 0.5)  | (0.2, 0.3, 0.4) | (0.15, 0.3, 0.5)  | (0.15, 0.3, 0.45) |
| Medium/Mid          | (0.35, 0.5, 0.65) | (0.4, 0.5, 0.6) | (0.3, 0.5, 0.7)   | (0.35, 0.5, 0.65) |
| High/Late           | (0.5, 0.65, 0.8)  | (0.6, 0.7, 0.8) | (0.5, 0.7, 0.85)  | (0.55, 0.7, 0.85) |
| Very High/Very Late | (0.7, 0.85, 1.0)  | (0.8, 0.9, 1.0) | (0.65, 0.85, 1.0) | (0.75, 1.0, 1.0)  |

### E. Fuzzy Rule Base

The rule base encodes how iteration progress and diversity map to inertia weight. The general idea is a standard one: high inertia early in the search or when the swarm has collapsed (low diversity), and low inertia later when the population still covers the space adequately.

The same rule semantics are preserved across all input set configurations and both granularity levels. Tables ?? and ?? present the complete rule bases.

TABLE III

FUZZY RULE BASE FOR INERTIA WEIGHT ADAPTATION (3 LABELS, 9 RULES).

| Rule | Progress | Diversity | Output $w$ |
|------|----------|-----------|------------|
| 1    | early    | low       | high       |
| 2    | early    | medium    | high       |
| 3    | early    | high      | high       |
| 4    | mid      | low       | low        |
| 5    | mid      | medium    | medium     |
| 6    | mid      | high      | high       |
| 7    | late     | low       | low        |
| 8    | late     | medium    | low        |
| 9    | late     | high      | medium     |

TABLE IV

FUZZY RULE BASE FOR INERTIA WEIGHT ADAPTATION (5 LABELS, 25 RULES).

| Rule | Progress   | Diversity | Output $w$ |
|------|------------|-----------|------------|
| 1    | very early | very low  | very high  |
| 2    | very early | low       | very high  |
| 3    | very early | medium    | high       |
| 4    | very early | high      | high       |
| 5    | very early | very high | medium     |
| 6    | early      | very low  | very high  |
| 7    | early      | low       | high       |
| 8    | early      | medium    | high       |
| 9    | early      | high      | medium     |
| 10   | early      | very high | medium     |
| 11   | mid        | very low  | high       |
| 12   | mid        | low       | high       |
| 13   | mid        | medium    | medium     |
| 14   | mid        | high      | medium     |
| 15   | mid        | very high | low        |
| 16   | late       | very low  | high       |
| 17   | late       | low       | medium     |
| 18   | late       | medium    | low        |
| 19   | late       | high      | low        |
| 20   | late       | very high | very low   |
| 21   | very late  | very low  | medium     |
| 22   | very late  | low       | low        |
| 23   | very late  | medium    | very low   |
| 24   | very late  | high      | very low   |
| 25   | very late  | very high | very low   |

## IV. EXPERIMENTAL SETUP

### A. Benchmark Functions

The experimental evaluation uses six classical benchmark functions representing two structurally distinct groups: high-dimensional problems with regular landscapes and low-dimensional problems with rugged, deceptive landscapes.

The first group includes three functions evaluated in  $d = 100$  dimensions. The Sphere function (F1) is a unimodal baseline with a smooth, convex landscape. Rosenbrock's function (F5) provides a unimodal challenge characterized by a narrow, curved valley where convergence is highly sensitive to search direction. Griewank's function (F11) introduces regular

multimodality through oscillatory components superimposed on a quadratic base.

The second group comprises three variants of the Shekel function in a four-dimensional space ( $d = 4$ ), differing in the number of components  $m$ : F21 ( $m=5$ ), F22 ( $m=7$ ), and F23 ( $m=10$ ). The Shekel family is characterized by closely spaced deep optima, making it particularly challenging for population-based methods prone to premature convergence.

All functions were evaluated through the `Opfunu` library [?], which provides reference implementations of clas-

sical benchmarks. Table ?? summarizes the selected benchmarks.

### B. Parameter Settings and Evaluation Protocol

All algorithmic variants were evaluated under identical conditions. The population size was fixed at 50 particles and the maximum number of iterations was set to 500. No additional stopping criteria were applied.

Each algorithm–function combination was executed over 31 independent runs. A shared seeding strategy was adopted: all algorithms used the same random seed for the  $k$ -th run, with the initial seed set to 42 and incremented by one for each subsequent run, yielding seeds  $\{42, 43, \dots, 72\}$ .

The baseline PSO with linearly decreasing inertia weight was evaluated alongside eight fuzzy-controlled variants (PSO–FCS). These variants result from the full factorial combination of the four input set configurations described in Section ?? (I1–I4) and two granularity levels (3 and 5 labels), with the output fuzzy set (A) and rule base held constant. Table ?? summarizes the nine configurations, totaling  $9 \times 6 \times 31 = 1,674$  independent experiments.

Performance was assessed using the final best fitness value and the relative gap to the known global optimum, defined as

$$\text{Gap}(\%) = \frac{|f_{\text{best}} - f_{\text{opt}}|}{|f_{\text{opt}}|} \times 100. \quad (6)$$

For functions with  $f_{\text{opt}} = 0$  (F1, F5, F11), performance is reported directly as the absolute best fitness value, since the relative gap is undefined.

Robustness was evaluated via the mean and standard deviation of final fitness values across the 31 runs. Statistical significance was assessed using the Wilcoxon signed-rank test at  $\alpha = 0.05$ , which exploits the paired design enabled by shared random seeds. Computational cost was measured using wall-clock execution time.

All experiments were conducted on a 64-bit system with an Intel Core i7 CPU (2.8 GHz) and 16 GB of RAM. The algorithms were implemented in Python 3.11.9 and executed sequentially.

## V. RESULTS AND DISCUSSION

The results fall into three parts and are discussed below: the overall comparison between baseline PSO and the fuzzy-controlled variants, the effect of input set geometry, and the role of linguistic granularity.

### A. Overall Performance: PSO vs PSO–FCS

Table ?? summarizes the comparison between the baseline PSO and the best-performing PSO–FCS variant for each benchmark function. All six functions are minimization problems: F1, F5, and F11 have a global optimum of zero, while the Shekel functions F21–F23 have negative global optima ( $-10.1532$ ,  $-10.4029$ , and  $-10.5363$ , respectively). The best fuzzy configuration per function was therefore selected by lowest mean fitness across 31 runs in all cases.

The results differ markedly between unimodal and multimodal landscapes. On F1, F5, and F11 the baseline PSO

achieves fitness values that are orders of magnitude better than any fuzzy variant ( $p < 0.001$ , Wilcoxon signed-rank test), with 20–28% less computation time. On unimodal landscapes, the single basin of attraction makes diversity-driven inertia adaptation unnecessary. The fuzzy controller reads the natural convergence of the swarm as low diversity and raises  $w$  to promote exploration—counterproductive when the swarm is already heading toward the global optimum.

On the Shekel functions (F21–F23), the situation is reversed. The best PSO–FCS variant improves mean fitness by 6.5–22.9% over the baseline and reduces standard deviation by 57–63%, with negligible overhead (1.0–5.3%). Here, the fuzzy controller detects premature convergence toward local optima and raises inertia to re-diversify the swarm. The 57–63% drop in standard deviation across 31 runs indicates that the controller not only improves average quality but also makes the search considerably more repeatable.

Statistical significance on the multimodal group is partial: only F21 reaches  $p < 0.05$  ( $p = 0.041$ ); F22 and F23 do not. The Shekel family has closely spaced deep optima, so small changes in initial positions can redirect the swarm to a different attractor, which inflates within-group variance and weakens the paired test.

Fig. ?? shows the fitness distributions for all six functions. On the Shekel group, fuzzy variants produce tighter interquartile ranges and fewer extreme outliers than the baseline, while on unimodal functions the gap in favor of standard PSO is visually evident.

Fig. ?? plots mean fitness against standard deviation for all nine configurations on each function, adding a second perspective on the same data. On the Shekel functions the PSO baseline (star marker) sits in the high-variance region, while the FCS variants group toward lower standard deviations. On unimodal functions the picture is the opposite: every FCS configuration lands far from the PSO baseline, well into the worse-fitness region.

### B. Effect of Input Set Geometry

To determine whether input partition geometry actually matters, Table ?? reports the mean rank of each input set across all six benchmark functions, computed separately per granularity level. A rank of 1 corresponds to the best configuration on a given function; a lower average rank signals more consistent performance across the suite.

With three labels, I1 (Standard) obtains the best average rank at 2.83. I2 (Narrow) comes second at 3.67; its sharper label boundaries seem to help on the Shekel functions, where the controller needs to react quickly to diversity drops. I3 (Wide) is the worst at both granularity levels (5.50 and 5.83). The heavy overlap between its labels appears to blur the control signal: the controller cannot tell apart states that, from a search perspective, are meaningfully different.

Fig. ?? presents the fitness heatmap for the multimodal functions, where the effect of input set geometry is most pronounced. I1 and I2 consistently achieve the darkest (best)

TABLE V  
BENCHMARK FUNCTIONS USED IN THE EXPERIMENTAL EVALUATION.

| Function              | Type       | Dim. | Bounds          | Global Opt. |
|-----------------------|------------|------|-----------------|-------------|
| F1 (Sphere)           | Unimodal   | 100  | $[-100, 100]^d$ | 0.0         |
| F5 (Rosenbrock)       | Unimodal   | 100  | $[-30, 30]^d$   | 0.0         |
| F11 (Griewank)        | Multimodal | 100  | $[-600, 600]^d$ | 0.0         |
| F21 (Shekel, $m=5$ )  | Multimodal | 4    | $[0, 10]^d$     | -10.1532    |
| F22 (Shekel, $m=7$ )  | Multimodal | 4    | $[0, 10]^d$     | -10.4029    |
| F23 (Shekel, $m=10$ ) | Multimodal | 4    | $[0, 10]^d$     | -10.5363    |

TABLE VI  
ALGORITHM CONFIGURATIONS EVALUATED. ALL PSO-FCS VARIANTS SHARE THE SAME RULE BASE AND OUTPUT SET; ONLY INPUT MEMBERSHIP FUNCTIONS AND GRANULARITY VARY.

| Algorithm      | $w$ Control     | Output Set | Labels | Input Set     |
|----------------|-----------------|------------|--------|---------------|
| PSO (baseline) | Linear decrease | —          | —      | —             |
| PSO-FCS-I1-3L  | Fuzzy (FCS)     | A          | 3      | I1 (Standard) |
| PSO-FCS-I2-3L  | Fuzzy (FCS)     | A          | 3      | I2 (Narrow)   |
| PSO-FCS-I3-3L  | Fuzzy (FCS)     | A          | 3      | I3 (Wide)     |
| PSO-FCS-I4-3L  | Fuzzy (FCS)     | A          | 3      | I4 (Shoulder) |
| PSO-FCS-I1-5L  | Fuzzy (FCS)     | A          | 5      | I1 (Standard) |
| PSO-FCS-I2-5L  | Fuzzy (FCS)     | A          | 5      | I2 (Narrow)   |
| PSO-FCS-I3-5L  | Fuzzy (FCS)     | A          | 5      | I3 (Wide)     |
| PSO-FCS-I4-5L  | Fuzzy (FCS)     | A          | 5      | I4 (Shoulder) |

TABLE VII  
PSO VS BEST PSO-FCS VARIANT PER FUNCTION. FITNESS IMPROVEMENT, STANDARD DEVIATION REDUCTION, WILCOXON SIGNED-RANK  $p$ -VALUE, AND COMPUTATIONAL OVERHEAD ARE REPORTED OVER 31 INDEPENDENT RUNS.

| Func. | Win | Best FCS | Impr.% | $\sigma$ Red.% | $p$       | OH%  |
|-------|-----|----------|--------|----------------|-----------|------|
| F1    | PSO | I1:3L    | —      | —              | <0.001*** | 20.1 |
| F5    | PSO | I1:5L    | —      | —              | <0.001*** | 27.6 |
| F11   | PSO | I1:3L    | —      | —              | <0.001*** | 20.2 |
| F21   | FCS | I2:5L    | +22.9  | -63.0          | 0.041*    | 3.6  |
| F22   | FCS | I1:3L    | +6.5   | -58.7          | 0.824 ns  | 5.3  |
| F23   | FCS | I2:3L    | +9.9   | -57.3          | 0.504 ns  | 1.0  |

TABLE VIII  
MEAN RANK OF EACH INPUT SET CONFIGURATION ACROSS SIX BENCHMARK FUNCTIONS (1 = BEST). RANKS COMPUTED INDEPENDENTLY PER GRANULARITY LEVEL.

| Input Set     | 3-Label Rank | 5-Label Rank |
|---------------|--------------|--------------|
| I1 (Standard) | <b>2.83</b>  | 4.50         |
| I2 (Narrow)   | 3.67         | 5.17         |
| I4 (Shoulder) | 4.50         | 4.00         |
| I3 (Wide)     | 5.50         | 5.83         |

cells, while I3 and I4 show lighter (worse) values, particularly on F21 and F22.

### C. Effect of Linguistic Granularity

Across the board, three-label partitions outrank their five-label counterparts. Table ?? shows that three labels yield a better mean rank for three of the four input sets (I1, I2, I3); only I4 (Shoulder) benefits marginally from the finer partition.

The average Gap% confirms the trend: 23.7% for three labels versus 24.1% for five labels. The absolute difference is

small, but the direction is the same across most configurations, pointing to over-segmentation of the input space when five labels are used.

A plausible explanation is that diversity and iteration progress, both bounded in  $[0, 1]$ , are coarse-grained signals by nature. Three labels per input already produce overlapping rule activations that smooth transient noise. Splitting the same domain into five labels creates narrower activation regions that amplify measurement noise instead of capturing genuinely different search states.

## VI. CONCLUSIONS

This work investigated the sensitivity of fuzzy-controlled PSO to the geometric design of input membership functions, a factor that has received limited explicit attention in the literature. By crossing four input set geometries (I1–I4) with two granularity levels (3 and 5 labels) under a constant output set and rule base, a total of 1,674 experiments were conducted across six benchmark functions with distinct landscape characteristics.

The experiments yield three observations.

First, changing the input membership functions does change controller performance by a non-negligible margin. On the Shekel functions the best fuzzy variant improved mean fitness by up to 22.9% and cut standard deviation by up to 63% relative to baseline PSO; on unimodal functions, PSO dominated every fuzzy configuration. Within the functions tested, this supports the view that diversity-driven inertia adaptation is more beneficial on landscapes with deceptive local optima than on smooth unimodal surfaces.

The second observation concerns partition geometry. Moderate overlap (I1) achieved the best average rank (2.83 at

figures/08\_pso\_vs\_fcs\_boxplot.png

Fig. 2. Fitness distributions (31 runs) for PSO and the best PSO-FCS variant per function. On unimodal functions (F1, F5, F11), PSO dominates. On multimodal functions (F21–F23), the fuzzy-controlled variant achieves better medians and reduced dispersion.

three labels), while excessive overlap (I3) ranked last at both granularity levels. Too much overlap blurs the boundary between search states and leads to an inertia trajectory that barely reacts to real changes in diversity or progress.

Finally, three-label partitions outperformed five-label partitions for three of the four input sets, with an average Gap% of 23.7% versus 24.1%. The coarser granularity produces more robust rule activation patterns on the two normalized inputs

used in this controller, while five labels over-segment the input space without providing additional useful discrimination.

From a design standpoint, input membership function geometry should be treated as an explicit variable rather than left at a default setting—especially when the controller inputs are low-resolution signals such as normalized diversity and iteration progress. As a practical guideline, moderate-overlap triangular partitions with three linguistic labels offer a reasonable starting

figures/04\_scatter\_fitness\_std.png

Fig. 3. Mean fitness vs. standard deviation for all configurations on each benchmark function. Colors encode input sets (I1–I4), markers distinguish granularity ( $\circ=3L$ ,  $\square=5L$ ), and the star denotes the PSO baseline. On multimodal functions, FCS variants achieve lower dispersion; on unimodal functions, all FCS variants are orders of magnitude worse than PSO.

point for fuzzy inertia weight controllers.

This study has limitations. Only triangular membership functions were tested; Gaussian or trapezoidal shapes may show different sensitivity patterns. The benchmark suite covers six functions and does not include high-dimensional multimodal problems, where the geometry–granularity interaction could behave differently. The output set and rule base were also held fixed; how they interact with the input partition is

still an open question.

Future work will extend this analysis in two directions: evaluating the interaction between input and output partition designs in a full factorial framework, and applying the methodology to combinatorial optimization problems where diversity dynamics differ substantially from continuous benchmarks.

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figures/01b\_heatmap\_multimodal.png

Fig. 4. Mean fitness heatmap for multimodal functions (F21–F23) across input sets and granularity levels. Darker cells indicate better fitness. I1 and I2 configurations consistently outperform I3 and I4.